



Internet of things technology

IOT mini-project

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Web-of-things: Smart Air Filter

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Abstract—In this article, design considerations, implementation, construction and demonstration of a low-cost embedded sensing platform, which can be integrated into a scaleable IOT telemetry architecture. Relying on industry standard technology such as MQTT, influxDB and Grafana for real-time environmental monitoring and visualization.

I. INTRODUCTION

As academics, we spend a lot of time setting in our rooms, not necessary worrying about our surroundings, not a presence context, but in a breatheable, healthy context. Many studies[LINK TIL DEM HER] show that spending time in negatively impacts health and might even go as far as affecting the way we think, indirectly affecting our minds. Many companies release IoT devices to monitor our bodily functions, such as heart rate, stress levels and insulin, and others have even released air purifiers, to combat this very problem. However, a common denominator for these solutions is that they are either very expensive or tie the user down in complicated software eco-systems, and often the cases is a combination of both.

With the increasing availability of low-cost embedded systems and wireless communication technologies, the adoption of IoT solutions has accelerated the application in environmental monitoring.

IoT based monitoring platforms can be implemented to use lightweight communication protocols and scaleable data-processing architectures. MQTT is one such protocol. Its low overhead and subscribe/publish communication model has made a widely adopted model for IoT applications. This makes it attractive when combining it with time-series databases such as InfluxDB and visualization platforms such as grafana.

This project presents a design and implementation of a IoT-enabled environmental monitoring and fan-control system based on a newer PIC16 microcontroller and a Rasperry PI gateway. The embedded subsystem acquires air quality measurements from an ENS160 gas sensor, while simultaneously monitoring multiple PWM-controlled fans. Telemetry is transmitted through a custom UART communications protocol to the Rasperry Pi, where data is published to MQTT, processing using Telegraf, stored in InfluxDb and visualized in Grafana dashboards.

The purpose of this paper is to demonstrate a low-cost, lightweight, embedded platform, that can be integrated into

larger, modular IoT architectures, for monitoring and Telemetry visualization. The remainder of this reports describes the theoretical background, system architecture, implementation details, experimental verification and future improvements.

II. BACKGROUND

A. MQTT communication protocol

Message queuing telemetry transport (*MQTT*) is a lightweight communication protocol, designed for transmissions between IoT systems. It implements a subscribe/publish model, in which devices communicate through a broker, instead of directly to one another. A device publish a specific topic, while other devices can subscribe to that particular topic. This enables increased scalability between distributed systems. This is a particular model is well suited for embedded solutions with low bandwidth requirements and minimal overhead, due to ie. power constraints or simple processor overhead.

In IoT environments MQTT topics are commonly organized into hierarchies in order to separate different types of data, whether it be measurements or other types of information.

B. Time-Series Databases and InfluxDB

Environmental monitoring systems continuously generates timestamped data. This data must be stored to create a log of both current and past behavior. Conventional relational databases are not optimized for this, due to ingestion speed at scale. For this task, time-series databases are designed to handle timestamped data efficiently.

InfluxDB is an open-source time-series database commonly used in IoT.

C. Grafana data visualization

visualization platforms are important tools in modern IoT systems. They provide intuitive access to telemetry gathered from embedded devices. And real-time dashboards allows users to observe system behavior. Grafana is an open-source visualization platform commonly used together with influxDB.

D. Embedded systems and microcontrollers

Embedded systems are smaller specialized computing systems designed to perform smaller tasks within larger electronic systems. The operation is deterministic, which results in predictable behavior and low-power consumption. Their main purpose is to interface to hardware, to bridge the communication to sensors, format data and pass it along.

microcontrollers also feature dedicated hardware, which can

be repurposed to control external hardware such as fans and do voltagem measurements.

E. Air quiality and ENS160 sensor

Indoor air quality monitoring has become increasingly important in residential, industrial and commercial environments. Poor air quality maybe negatively impact comfort, productivity and health. This is a partularly apparent in enclosed spaces. The ENS160 is a digital gas sensor made by ScioSense [INDSÆT LINK].

III. SYSTEM DESIGN AND IMPLEMENTATION

A. Overall System Architecture

The system proposed in the paper is designed as a modular IoT-based Environmental monitoring platform, consisting of an embedded sensing processor subsystem and a higher-level telemetry processor. The embedded subsystem is responsible for sensor data acquisition, fan control and monitoring, aswell as low-level hardware interaction.

The sensing subsystem is based on a PIC16F18124[INDSÆT REF] microcontroller, with its own auxillary supply, connected to an ENS160 air-quality sensor and multiple PWM-controlled fans. The microcontroller continuously acquires environmental measurements and fan telemetry while simultaneously regulating the fan speed. System information is transmitted to the Raspberry PI through the UART communication interface.

The Raspberry Pi acts a edge gateway between the embedded hardware and the IoT infrastructure. Incoming UART telemetry is processed by a python application and published to MQTT topics. This telemetry is subsequently collected by telegraf to be stored in an InfluxDB time-series databased, later to be visualized by grafana. It is the handling done by telegraf that timestamps the data, as the microcontroller system, effectively can't determine absolute time, only that which have passed since boot.

B. PIC Firmware architecture

IV. RESULTS

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V. CONCLUSION

conclusion

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